UNIVERSITY NAME

Analysis IV — Final Exam

Year: 2023

Exercise 1:

Determine the nature (convergence/divergence) of the following numerical series:

$$1)\ u_n = \frac{n^n}{n!}$$
$$2)\ u_n = \log\left(1 + \frac{2}{n(n+1)}\right)$$
$$3)\ u_n = \sin\left(\frac{\pi}{2}\frac{an+1}{\sqrt{n^2+an+1}}\right)$$

Answer Area

Exercise 2:

For all $n \in \mathbb{N}^*$, define the function:

$$f_n : \mathbb{R} \longrightarrow \mathbb{R}, \quad x \mapsto \frac{n^\alpha x}{1+n^2x^2}$$

When the series $\sum f_n(x)$ converges, denote $f_\alpha(x) = \sum_{n=1}^{\infty} f_n(x)$.

1. Determine, according to the values of α, the domain D_α of definition of f_α.

   From now on, assume $\alpha < 1$.

2. (a) Show that for all $x \in \mathbb{R}_+^*$,

   $$\frac{x}{1+x^2}n^{\alpha-2} \le f_n(x) \le \frac{1}{x}n^{\alpha-2}.$$

   (b) Show that there exists $A > 0$ (depending on α) such that

   $$\forall x \in \mathbb{R}_+^*, \quad \frac{Ax}{1+x^2} \le f_\alpha(x) \le \frac{A}{x}.$$

   Deduce an equivalent of f_α at ∞.

3. If $\alpha < 0$, show that

   $$\int_0^1 f_\alpha(t)\,dt = \sum_{n=1}^{\infty} \frac{1}{2}n^{\alpha-2}\log(1+n^2).$$

4. Assume $0 \leq \alpha < 1$.
   (a) Show that f_α is continuous on $\mathbb{R}^+$.
   (b) Show that for all $N \in \mathbb{N}^*$

$$f_\alpha\left(\frac{1}{N}\right) \geq N \sum_{n=1}^{N} \frac{n^\alpha}{N^2 + n^2}$$

   Deduce the discontinuity of f_α at 0.
   (c) Does the series $\sum f_n$ converge uniformly on $\mathbb{R}_+$?
5. (Optional) Show that f_α is differentiable on $(0, +\infty)$ and decreasing on $[1, +\infty)$.

Answer Area

Exercise 3:

Consider the differential equation:

$$4xy''(x) + 2y'(x) - y(x) = 0 \quad (E)$$

Let $\sum a_n x^n$ be a power series with radius of convergence R and sum f.

1. Justify that f is of class C^2 on $(-R, R)$. Deduce, for $x \in (-R, R)$, an expression of $f'(x)$ and $f''(x)$.
2. Show that f is a solution of (E) if and only if

$$a_{n+1} = \frac{a_n}{(2n+1)(2n+2)}, \qquad \forall n \in \mathbb{N}.$$

3. Express a_n in terms of n and a_0. Determine the radius of convergence of the power series $\sum a_n x^n$.
4. Expand into a power series the function defined by:

$$\varphi(x) = \begin{cases} \cosh(\sqrt{x}) & \text{if } x > 0 \\ 1 & \text{if } x = 0 \\ \cos(\sqrt{-x}) & \text{if } x < 0 \end{cases}$$

Answer Area